



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

SE 423: Introduction to Mechatronics

Lecture 20: Sensor Fusion and Kalman Filters

Marius Juston

Monday, April 6th, 2026

Talk to the person next to you and answer these questions.

- What assumptions were made during path planning / trajectory planning?
- What sensors on the robot help determine it's the location in the world?
- When could the odometry and the IMU give different values? Which one should you trust?
- If your IMU says you accelerated forward, but LiDAR scan matching says you didn't move, what should the system believe?

Office Hours:

- **Monday:** 4-6 PM, Neel
- **Tuesday:** 11-1 PM, Lakshmi
- **Tuesday:** 1-3 PM, Samuel
- **Tuesday:** 2-5 PM, Dan & Marius

Lab: Finishing Lab 6 this week. This a 3-week lab!

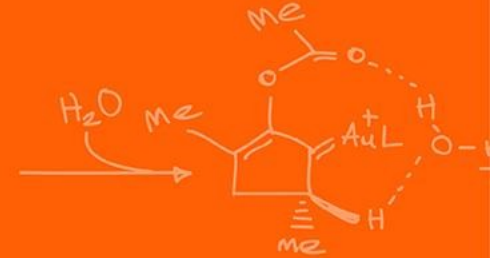
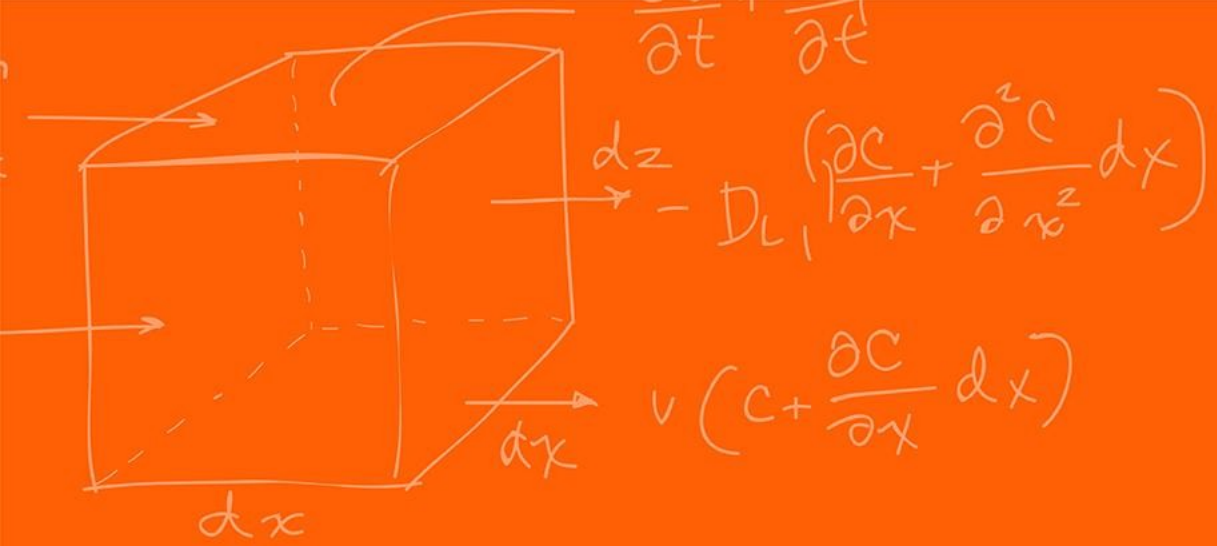
- Don't forget to have the Lab LabVIEW application implemented before lab.

Homeworks:

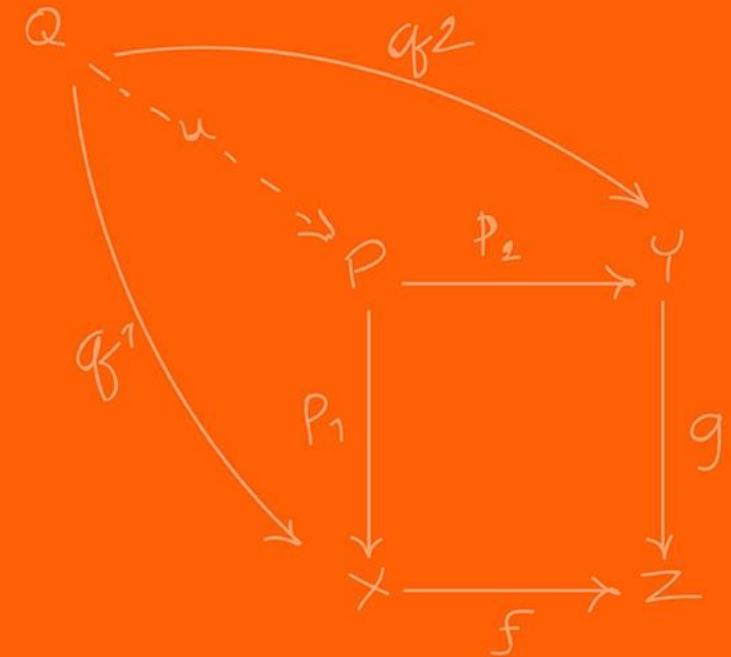
- Check-offs for [homework 4](#) Ex 4 are due **April 7, 5PM (The end of office hours)**
- Remainder of [homework 4](#) due on Gradescope on **April 8, 9AM**
- Next homework is primarily the **personal project** due on **April 22, 9AM**.
 - I would highly recommend you thinking now and maybe even starting it now if you have a good idea already!

Questions?

Homework, Lab, LabVIEW?



Sensor Fusion



Last lectures, we covered algorithms like A*, Dijkstras, Hybrid A* for path planning.

We learned to compute an optimal, collision-free trajectory from Point A to Point B.

However, path planning operates in a perfect, simulated world!

The **fundamental assumption** we made about the robot to execute these plans were:

Perfect localization and actuation!

In the real world, motors don't spin exactly as commanded due to friction and voltage drops.

Sensors have electrical noise, mechanical imperfections, and environmental interference.

What can cause the LiDAR to fail?

- Reflective / absorptive surfaces: glass, mirrors, black materials
- Fog, rain, dust, snow
- Fast motion relative to scan frequency

What can cause the odometry to fail?

- Slippery terrain for wheel slip

When can cause the IMU fail?

- IMU drift
- Saturating the sensors

If we blindly execute our path plan open-loop, the robot will quickly crash.

We need **Localization**: determining the robot's state (position, velocity, orientation) in the real world.

To solve the localization problem, SE 423 robots use multiple sensors.

Proprioceptive sensors measure internal states: Wheel Encoders (Odometry) and IMUs.

Exteroceptive sensors measure the external environment: LiDAR.

No single sensor is perfect; they all lie to us in different ways.

Wheel encoders count the rotations of the robot's wheels. By knowing wheel radius and track width, we calculate forward kinematics to estimate position. It provides **high-frequency**, smooth data for short-term movement.

Similarly, **IMUs** generate **high-frequency**, smooth data for short-term movement based on the integration of acceleration.

LiDAR, unlike odometry and IMUs, LiDAR gives us **absolute** measurements relative to the environment.; however, due to the additional data, the sensor runs at a lower frequency (e.g., 10Hz) compared to an IMU (e.g., 500Hz).

If we are in a perfectly square, featureless hallway, can LiDAR tell us our position along the hallway? What about the IMU / odometry?

We have **fast, smooth** sensors that drift (IMU, Odometry).

We have **slow**, computationally heavy sensors that don't drift (LiDAR).

How do we combine them?

Simple averaging doesn't work because their reliability changes over time.

If 2 sensors given measurements give different values which, should you trust?

What if your sensors give you the confidence of their measurements? I.e. the LiDAR is 75% sure of its position and the IMU / Odometry are, in total, 25% sure of their positions?

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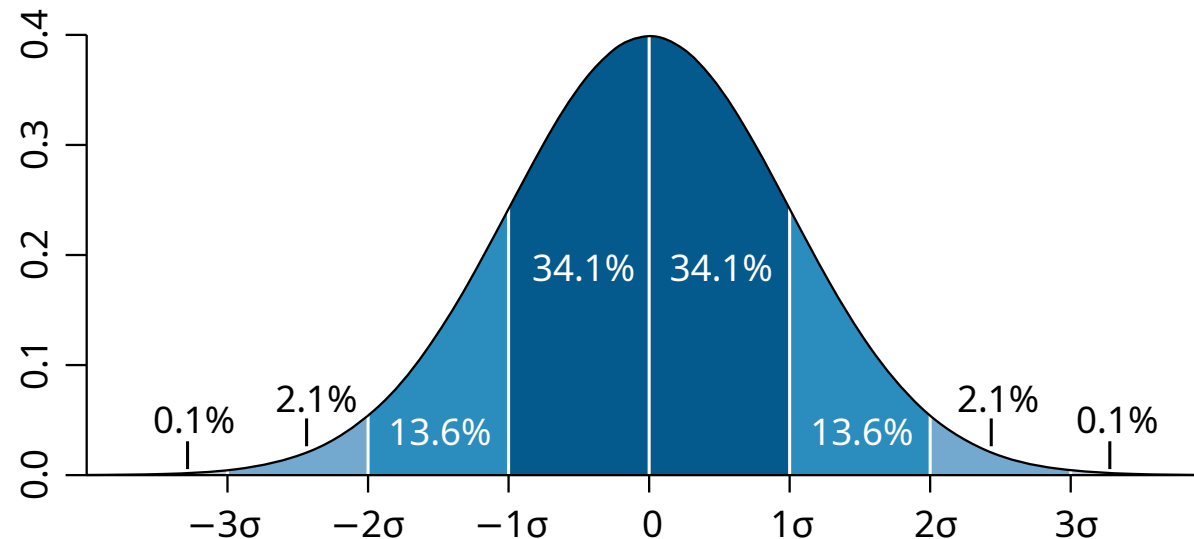
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What if your sensors give you the confidence of their measurements? I.e. the LiDAR is 75% sure of its position and the IMU / Odometry are, in total, 25% sure of their positions?

Instead of saying robot is exactly at “ $X = 5$ ”, we want to say “We are 90% sure that the robot is near $X=5$ ”.

We represent our state as a probability distribution.

We use the Gaussian (Normal) Distribution, defined by a mean (μ) and variance (σ^2).



https://en.wikipedia.org/wiki/Normal_distribution

Why Gaussians?

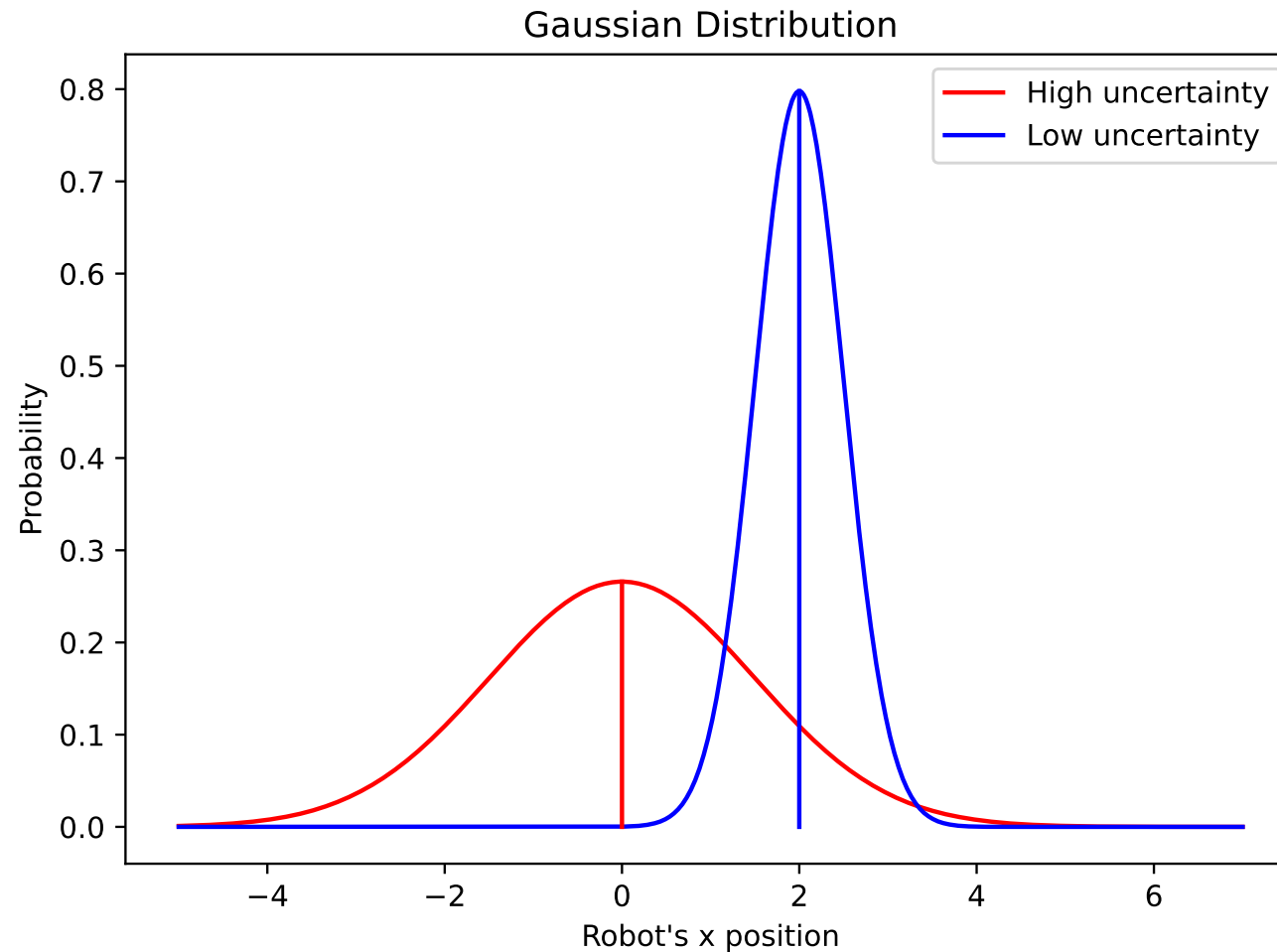
The **mean** (μ) represents our best guess of the robot's state.

The **variance** (σ^2) represents our uncertainty; a wide curve means we are lost, a narrow curve means we are confident.

Gaussians are mathematically convenient: multiplying two Gaussians results in a new Gaussian!

In addition, a lot of math can be derived in closed form when using Gaussians (most other more complex probability distributions do not allow for this)

Which sensor would you trust more?

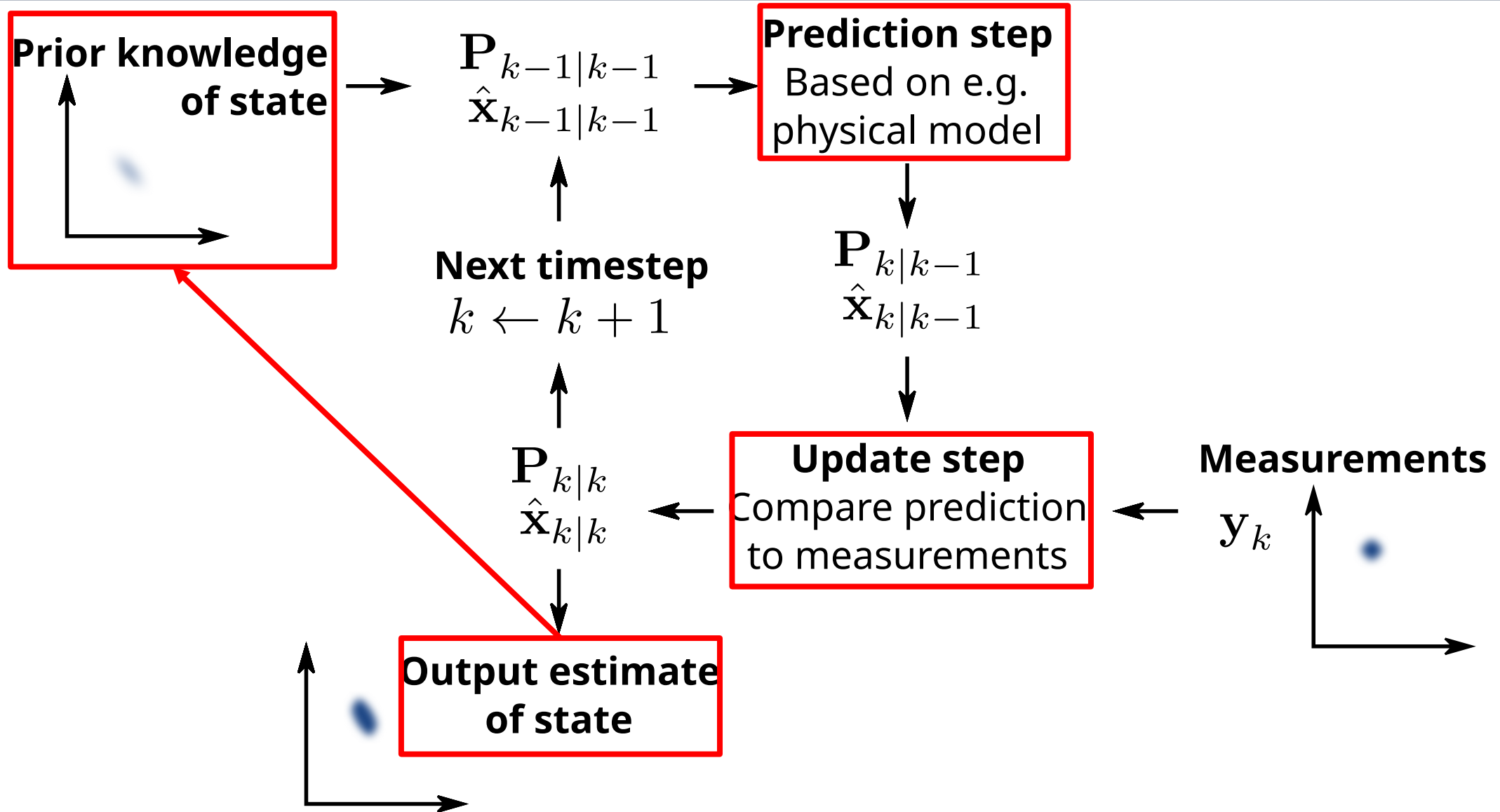


The Kalman Filter is an optimal estimator for **linear** systems subject to **Gaussian** noise.

When would the Kalman Filter not be appropriate?

If your sensor or system is non-linear and / or does not follow a Gaussian noise distribution! Always understand how your sensors / system acts and the underlying assumptions for algorithms!

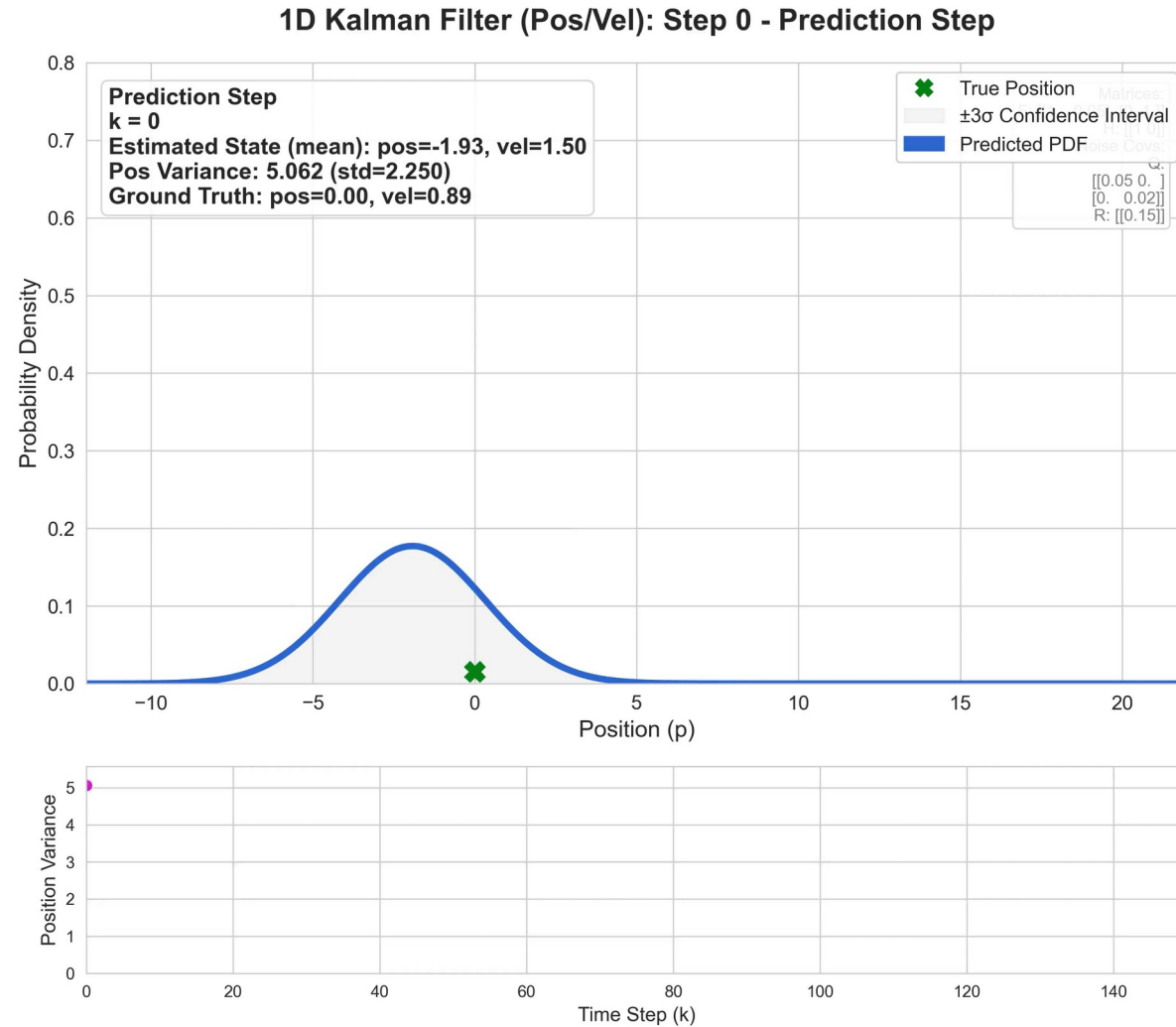
It operates in a continuous, two-step loop: **Predict** and **Update**.



Step 1: The robot **moves**. We use our fast, noisy sensors (Odometry/IMU) to **Predict** where we are. Our uncertainty grows.

Step 2: The robot **looks**. We take a measurement with our absolute sensor (LiDAR) to **Update** our belief. Our uncertainty shrinks.

Rinse and repeat thousands of times a second.



We use a physical model of the robot (kinematics) and our control inputs (u) to guess the new state.

Because we know our model and our encoders are imperfect, we actively increase our uncertainty covariance (P).

This represents "flying blind" between reference measurements.

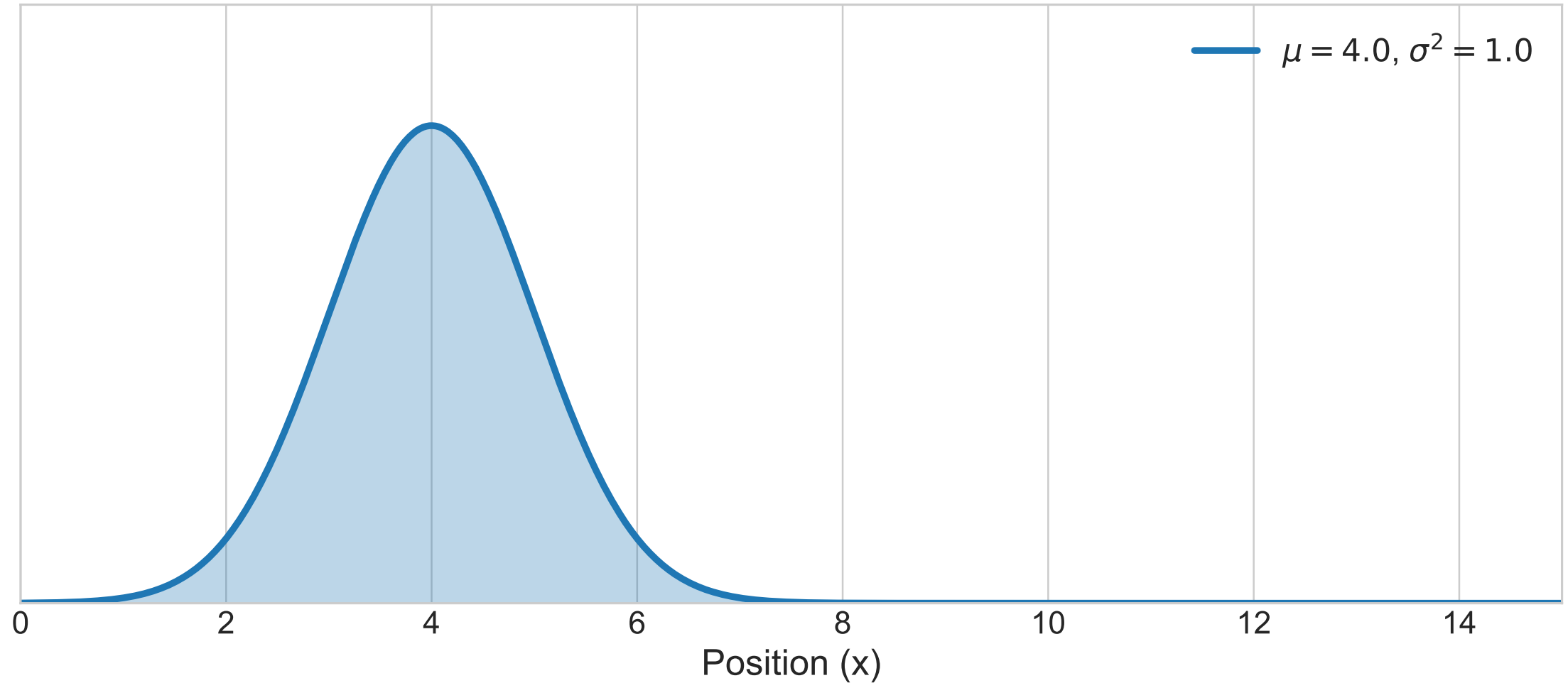
If we close our eyes and take 5 steps, are we more or less certain of our position than before we moved?

Before we look at massive matrices, let's look at a robot on a 1-Dimensional train track.

Its state is just one number: its position on the track, μ . Its uncertainty is also just one number: the variance, σ^2 .

Together, they form a 1D Gaussian (Bell Curve) representing our "belief".

Initial Belief



The robot commands its motors to move forward by distance u .

Our new predicted mean simply shifts:

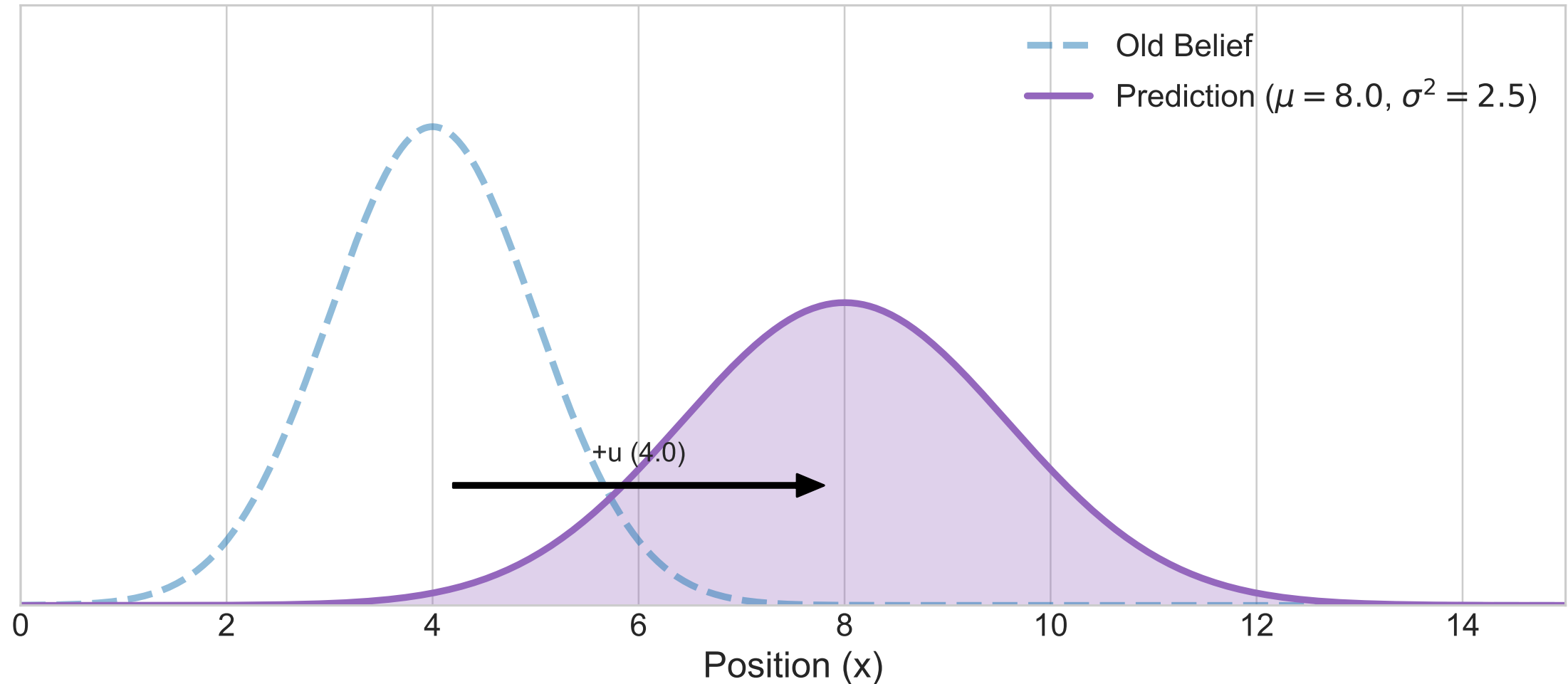
$$\mu_{pred} = \mu_{old} + u$$

But moving adds process noise (σ_{move}^2).

Our uncertainty grows.

$$\sigma_{pred}^2 = \sigma_{old}^2 + \sigma_{move}^2$$

1D Prediction (Shift & Spread)



This prediction updates the covariance and the means as such because we are assuming **Gaussian** random variables.

Let X and Y be independent random variables that are normally distributed, then their sum is also normally distributed. i.e., if

$$X \sim N(\mu_X, \sigma_X^2), \quad Y \sim N(\mu_Y, \sigma_Y^2) \\ Z = X + Y$$

Then,

$$Z \sim N(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2)$$

For how to derive it you can look [here \(from Wikipedia\)](#)

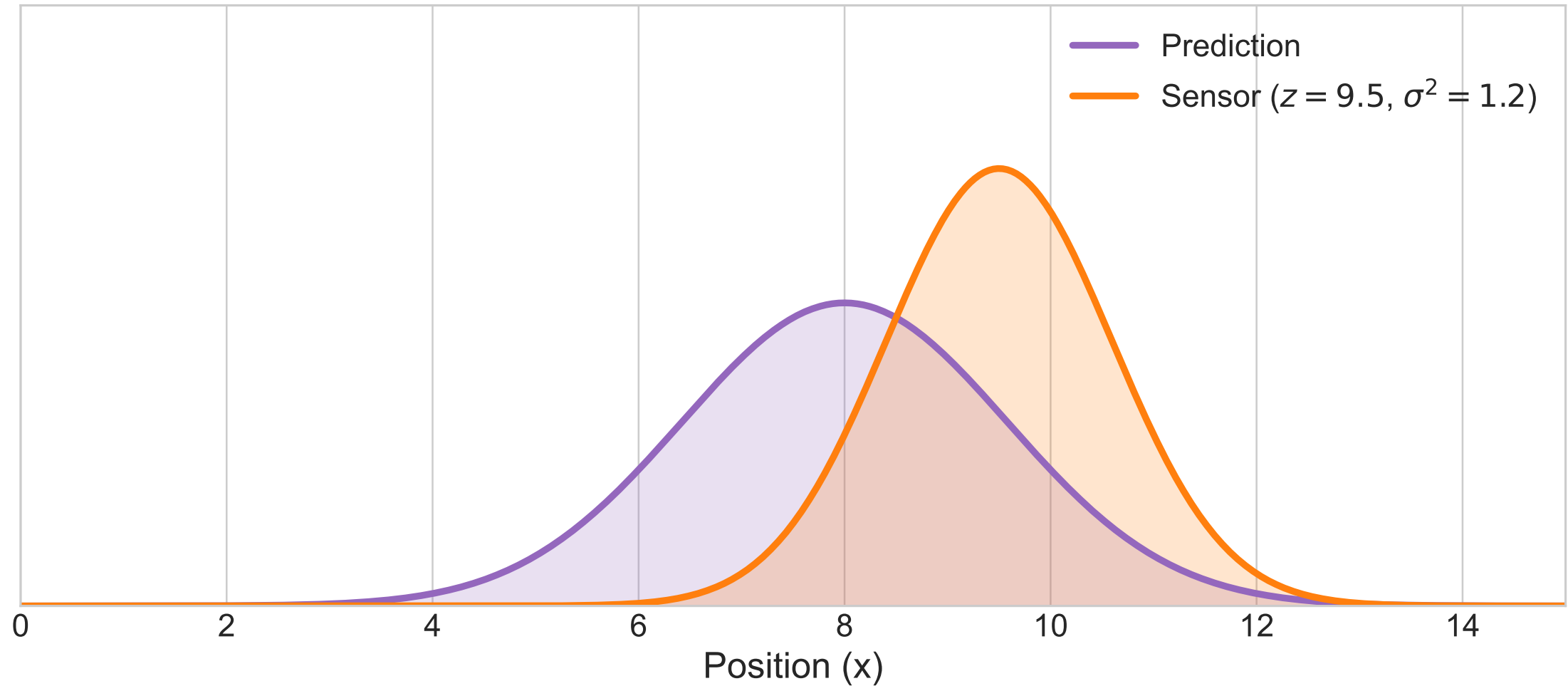
Now, our absolute sensor (like a 1D LiDAR) **takes a measurement**.

It says the robot is at position z , with a sensor noise variance of σ_{sensor}^2 .

We now have **two** Gaussian curves: our flattened **prediction** curve, and the new **sensor** curve.

How do we combine these two conflicting opinions into one single truth?

1D Update (Conflicting Opinions)



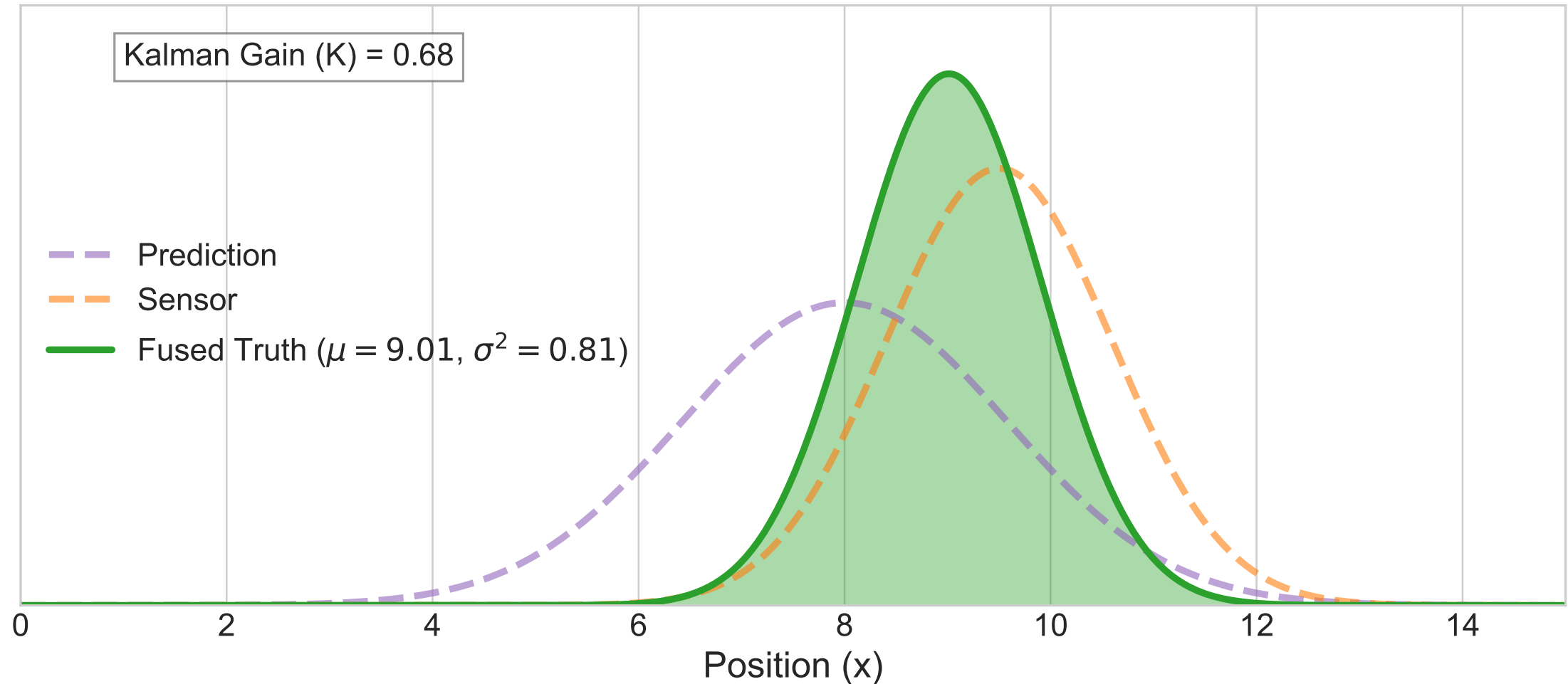
In probability, to fuse two independent estimates, we **multiply their distributions** together.

The **magical property** of Gaussians: Multiplying two Gaussians **always** produces a new **Gaussian**.

Because the new curve represents the overlapping area of agreement, it is always taller and narrower than both original curves.

Fusing data **reduces** uncertainty!

Fusing Data (Multiplying Gaussians)



The math behind that multiplication gives us the 1D **Kalman Gain** (K):

$$K = \frac{\sigma_{pred}^2}{\sigma_{pred}^2 + \sigma_{sensor}^2}$$

K is just a ratio! It's the ratio of our prediction uncertainty to the total system uncertainty. It always falls between 0 and 1.

We update our position using this ratio:

$$\mu_{new} = \mu_{pred} + K(z - \mu_{pred})$$

All it really says is “how much of the measurement should we incorporate”?

That's all the Kalman filter is!

What is the current limitation of the formulation?

We are only looking at a problem with 1 state, can this be generalized to be N-dimensional?

For our differentiable robot, we live in a 2D plane with an X , Y , and heading θ .

We can no longer use a single scalar number μ for position. We must use a **State Vector**: $x = [X, Y, \theta]^T$.

We can no longer use a single scalar variance σ^2 . We must use a **Covariance Matrix**, P .

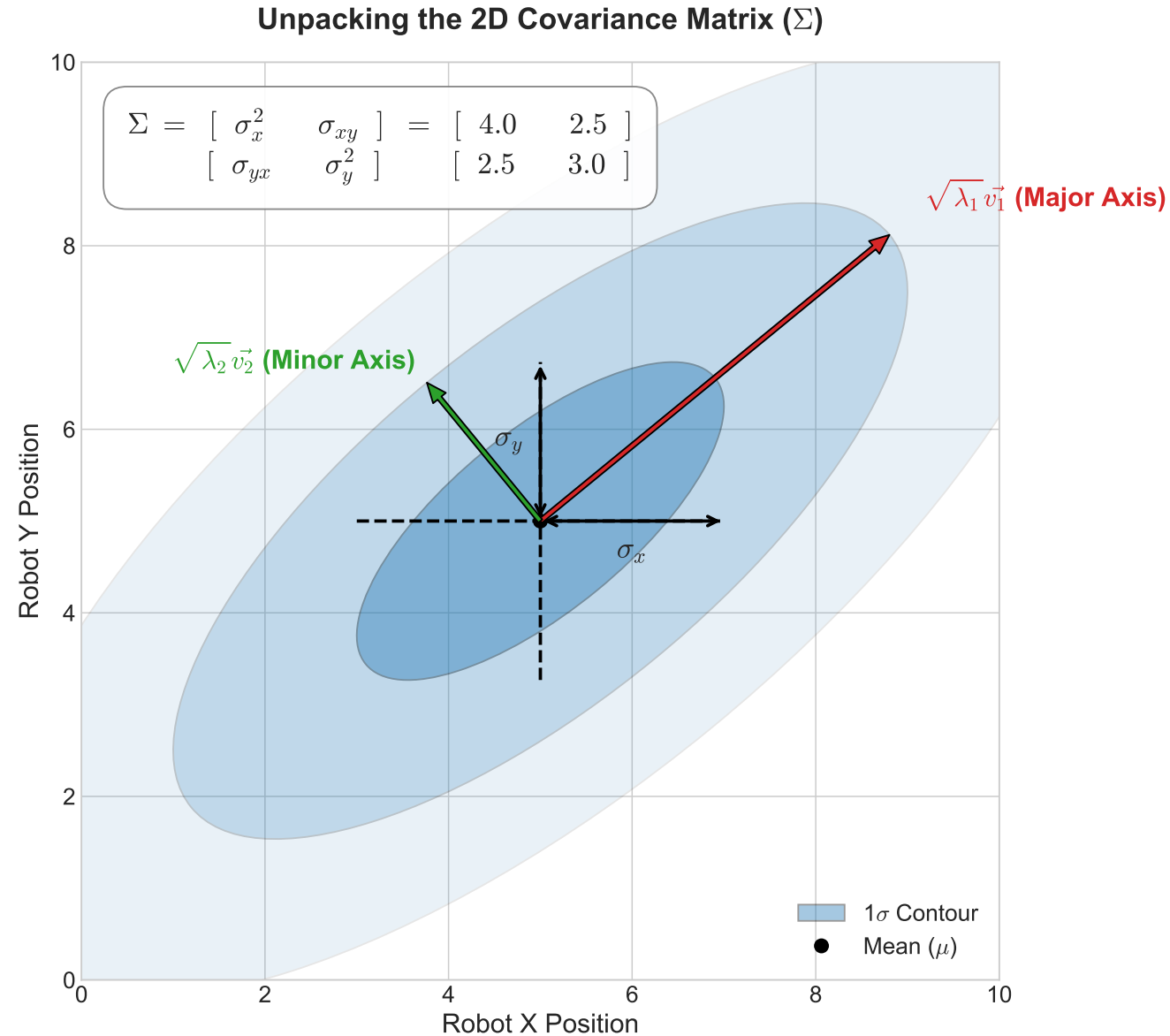
The **Covariance Matrix** P represents multi-dimensional uncertainty as an ellipse, not a 1D curve.

The diagonal elements are the individual variances (e.g., how unsure are we about X independent of everything else?).

The off-diagonal elements are the critical addition: **Correlations**.

If our robot only drives forward, and we are uncertain about our heading θ , will that create an uncertainty in Y as we travel?

Yes, errors in θ are highly correlated with errors in X and Y .



Let's translate our 1D prediction ($\mu_{pred} = \mu_{old} + u$) to N-Dimensions:

$$x_{k|k-1} = Fx_{k-1|k-1} + Bu_k$$

F is our state transition matrix (how physics updates the state).

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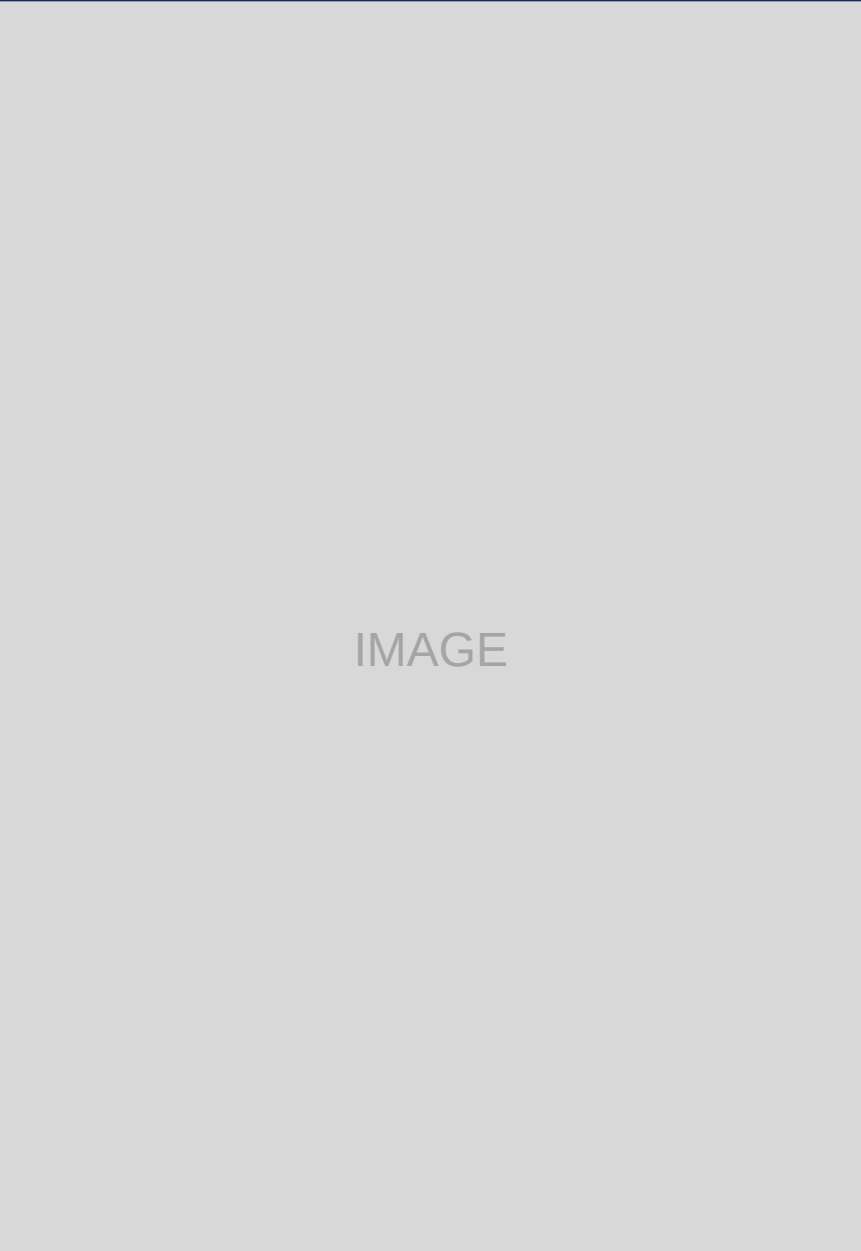


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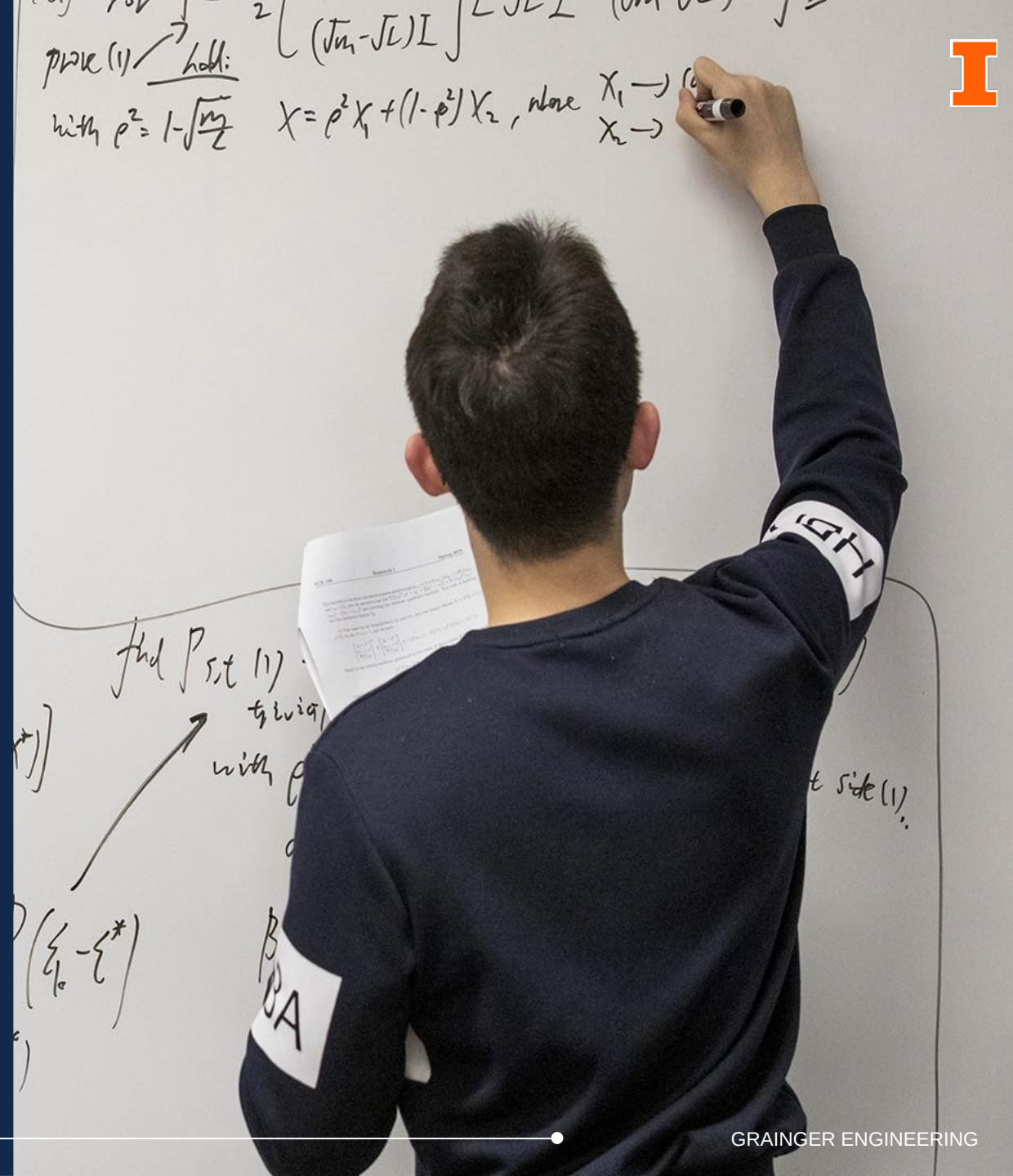
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Summary

Thank You

Questions

Contact Information

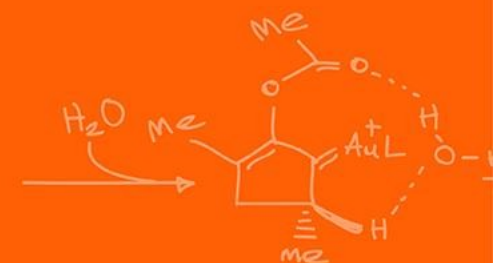
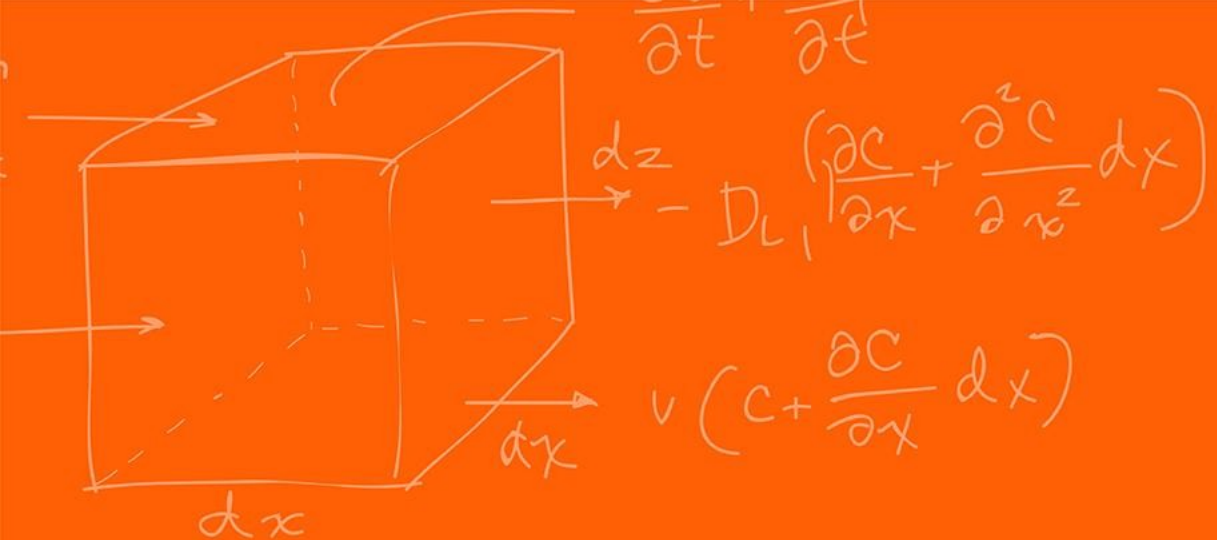
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Questions?



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